

Cairo Branch – Network Configuration Walkthrough

Internal Routing (EIGRP) & External Routing (BGP) via Service Provider

Project Scope: Implementing internal routing (EIGRP) for local departments and external routing (BGP) for inter-branch communication via a Service Provider.

1. Internal Branch Router (Router 1 — same logic applies to Router 2)

Role: Connects the local departments (Sales & HR) to the main Cairo hub (R3) using EIGRP. Router 2 follows this exact same logic for the IT department.

A Interface Configuration — Assigning IP Addresses to Physical Ports

eth0 and eth1 connect to the department switches; eth2 connects to the R3 hub using a /30 point-to-point link.

```
frr# conf t
frr(config)# interface eth0
frr(config-if)# ip address 20.1.1.1/24
frr(config-if)# no shutdown
frr(config-if)# exit

frr(config)# interface eth2
frr(config-if)# ip address 10.1.1.1/30
frr(config-if)# no shutdown
frr(config-if)# exit
```

B EIGRP Configuration — Activating EIGRP (AS 100)

Activates EIGRP Autonomous System 100 so the router advertises its directly connected networks to the rest of the Cairo branch.

```
frr(config)# router eigrp 100
frr(config-router)# network 20.1.1.0/24
frr(config-router)# network 10.1.1.0/30
frr(config-router)# do write
```

2. The Border Gateway (Router 3)

Role: Acts as the central hub for Cairo. Speaks EIGRP to the internal routers (R1, R2) and BGP to the outside world (SP Cloud / Alexandria branch).

A External Interface Configuration — Connecting to the Service Provider

Prepares interface eth2, which physically connects R3 to the Service Provider Cloud.

```
frr# conf t
frr(config)# interface eth2
frr(config-if)# ip address 10.1.3.1/30
frr(config-if)# no shutdown
frr(config-if)# exit
```

B BGP Configuration & EIGRP Redistribution into BGP

Establishes the external BGP peering with the Alexandria branch (AS 200) and injects Cairo's internal EIGRP routes into the BGP process so they are reachable externally.

```
frr(config)# router bgp 100
frr(config-router)# neighbor 10.1.3.2 remote-as 200
frr(config-router)# address-family ipv4 unicast
frr(config-router-af)# redistribute eigrp
frr(config-router-af)# exit
frr(config-router)# exit
```

C BGP Redistribution into EIGRP — Sharing External Routes Internally

Takes routes learned from BGP (the outside world) and shares them back into the internal Cairo EIGRP domain so all internal routers know how to reach Alexandria.

```
frr(config)# router eigrp 100
frr(config-router)# redistribute bgp
frr(config-router)# do write
```

3. End Devices (Example: VPCS PC6 — Sales Department)

Role: An end-user workstation in the Sales department.

A Assigning IP Address and Default Gateway

Sets the host IP, subnet mask (CIDR notation), and points the workstation to R1 as its default gateway.

```
PC6> ip 20.1.1.2 /24 20.1.1.1
```

B Saving the Configuration

Ensures the PC retains its IP address after the GNS3 simulator is restarted.

```
PC6> save
```

4. Key Verification Commands

These commands were used at various stages to confirm network reachability and protocol status.

Command	Run On	Purpose
show ip route	R1 / R2 / R3	Displays the routing table. Routes marked D are EIGRP; routes marked D EX are External EIGRP (redistributed from BGP).
show ip bgp summary	R3	Shows BGP neighbor handshake status. Active = attempting to connect; a numerical value = Established connection.
ping [IP Address]	PCs / Routers	Tests end-to-end ICMP reachability to verify the data plane is functioning across all configured routing protocols.

Cairo Branch Network Configuration | EIGRP AS 100 · BGP AS 100 → AS 200 | Topology: R1 & R2 (internal) → R3 (border) → SP Cloud → Alexandria